Student Academic Performance Risk Prediction Using Machine Learning

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Abstract

This project builds a small end-to-end machine learning application that estimates whether a student may be at risk of low academic performance. A student is labelled At Risk if the final grade G3 is below 10 out of 20, a threshold defined for this project. Using the UCI Student Performance dataset (395 students), two classifiers were trained inside scikit-learn Pipelines: Logistic Regression as a baseline and a Random Forest for comparison. On a held-out test set of 79 students, the Random Forest reached a precision of 0.85, a recall of 0.88 and an F1-score of 0.87 for the At Risk class, compared with 0.81, 0.85 and 0.83 for Logistic Regression. The models are delivered through a Streamlit dashboard with input validation, charts, an evaluation view and a responsible-use disclaimer. The output is a decision-support indicator to guide supportive review, not a judgment about any student.

1. Introduction and Problem Statement

Schools and colleges often hold attendance, previous grades, study-time, absence and assignment information, but reviewing every record manually is difficult. This project learns patterns associated with a defined low-performance outcome and flags records that may benefit from additional support.

The proposed output is a label of At Risk or Not At Risk, together with a short explanation of the risk definition and the input factors used.

2. Objectives

1. Define a clear and documented at-risk label from the available grades.
2. Inspect, clean and prepare the data, and avoid data leakage.
3. Train a baseline model (Logistic Regression) and a second model (Random Forest) and compare them using precision, recall, F1-score and the confusion matrix.

4. Build an interactive dashboard that accepts student features and shows the predicted category, relevant factors, charts, evaluation results and a disclaimer.
5. Test edge cases, validate inputs, and document limitations and responsible use.

3. Tools and Technologies

Tool or technology	Purpose
Python	Main programming language
VS Code	Development environment
Pandas, NumPy	Data cleaning and numerical processing
Scikit-learn	Preprocessing, training and evaluation
Logistic Regression	Baseline classification model
Random Forest Classifier	Second model for comparison
Matplotlib, Seaborn	Graphs and performance visualisation
Streamlit	Interactive dashboard
UCI Student Performance Dataset	Starter dataset

4. Dataset and Target Definition

The starter dataset is the UCI Student Performance Dataset, using the mathematics file (student-mat.csv). It contains 395 students and 33 columns covering student and school-related attributes, study habits, absences and three grades: G1 (period 1), G2 (period 2) and G3 (final), each out of 20.

The dataset does not directly include an at-risk label, so one was defined explicitly: a student is At Risk if the final grade G3 is below 10. This threshold is a project-defined assumption, not a universal standard.

Class	Students	Share
Not At Risk	265	67%
At Risk	130	33%

The classes are mildly imbalanced, which is why precision, recall and F1-score are reported and accuracy is not relied on alone.

5. Data Inspection and Cleaning

The data was inspected before modelling. The checks and results are summarised below.

Check	Result
Shape	395 rows and 33 columns
Data types	17 text columns and 16 numeric (integer) columns
Missing values	0
Duplicate rows	0
Text columns	Trimmed of extra spaces; all category values checked and no typos or unexpected values found
Grade ranges	G1, G2 and G3 all within 0 to 20
Age and absences	Age 15 to 22; absences 0 to 75
Absence outliers	15 students with absences above 20 (IQR rule)
Students with G3 = 0	38

The absence outliers and the students with a final grade of 0 were kept on purpose. They are real students who are likely to be at risk, and removing them would hide the exact cases the model is meant to detect. Because the data was already clean, no rows were removed. The cleaned file (student-mat-cleaned.csv) is the input to the modelling pipeline.

6. Exploratory Data Analysis

The table shows the average value of selected features for each group.

Feature	Not At Risk	At Risk
Study time (1 to 4)	2.08	1.95
Past failures	0.16	0.69
Absences	5.19	6.76
G1	12.45	7.76
G2	12.62	6.82

At-risk students have more past failures, more absences and clearly lower period 1 and period 2 grades.

[Insert figure: figures/eda_charts.png]

Figure 1: Final grade (G3) distribution with the risk threshold, absences by risk group, and class balance.

7. Feature Preparation and Data Leakage

Only features that would be available at the time of prediction were used. The model is designed to predict after the second-period exam, so G1 and G2 are valid inputs. The final grade G3 is used only to create the label and is never an input feature.

To reduce the use of unnecessary sensitive attributes, sex and romantic relationship status were removed. After these changes the model uses 30 input features: 15 numeric columns and 15 categorical columns. Categorical columns are one-hot encoded and numeric columns are standardised inside the modelling Pipeline, so the same preprocessing is applied at training and prediction time.

8. Methodology

1. The data was split 80/20 with stratification on the target and a fixed random_state of 42, giving 316 training students and 79 test students. The test set was kept separate until final evaluation.

2. A scikit-learn Pipeline combines a ColumnTransformer (StandardScaler for numeric columns, OneHotEncoder for categorical columns) with the classifier.
3. Logistic Regression (max_iter = 1000, balanced class weights) was trained as the baseline.
4. A Random Forest (200 trees, balanced class weights, random_state = 42) was trained for comparison.
5. Both models were evaluated on the held-out test set using precision, recall, F1-score and the confusion matrix, and were saved for use in the dashboard.

9. Results

Results for the At Risk class on the test set of 79 students (26 At Risk and 53 Not At Risk):

Model	Precision	Recall	F1-score	Accuracy
Logistic Regression	0.81	0.85	0.83	0.89
Random Forest	0.85	0.88	0.87	0.91

For the Not At Risk class, Logistic Regression scored precision 0.92, recall 0.91 and F1 0.91, and Random Forest scored 0.94, 0.92 and 0.93.

[Insert figure: figures/confusion_matrices.png]

Figure 2: Confusion matrices for Logistic Regression and Random Forest on the test set.

The Random Forest performed slightly better than the baseline on every metric. Recall for the At Risk class is especially important, because missing a student who needs support is the costly mistake. However, the test set contains only 79 students, so the difference between the two models is small and could change with a different split.

Feature importance

The Random Forest relies mostly on the period 2 grade (G2), followed by the period 1 grade (G1), then absences, past failures, going out with friends, age and health. In practice the model largely learns that low earlier grades predict a low final grade. This importance reflects what the model uses across all students and is not a full explanation of any single prediction.

[Insert screenshot: screenshots/05_feature_importance.png]

Figure 3: Factors the Random Forest relies on most.

10. Dashboard

The Streamlit dashboard has four tabs:

- **Predict:** an input form for academic features, a choice of model, the predicted category (At Risk or Not At Risk) with an estimated probability, the factors the model relies on, and a disclaimer.
- **Data charts:** grade distribution, absences by risk group and class balance.
- **Model evaluation:** precision, recall and F1-score for both models and their confusion matrices.
- **About and limitations:** the risk definition, how leakage was avoided, and responsible-use notes.

[Insert screenshot: screenshots/01_predict_at_risk.png]

Figure 4: Prediction screen showing an At Risk result with the disclaimer.

[Insert screenshot: screenshots/02_predict_not_at_risk.png]

Figure 5: Prediction screen showing a Not At Risk result.

[Insert screenshot: screenshots/06_data_charts.png]

Figure 6: Data charts tab.

[Insert screenshot: screenshots/07_model_evaluation.png]

Figure 7: Model evaluation tab.

[Insert screenshot: screenshots/08_about.png]

Figure 8: About and limitations tab.

11. Testing and Input Validation

The dashboard validates inputs before predicting:

- If both G1 and G2 are 0, an error is shown and no prediction is made.
- If G1 and G2 differ by more than 8 points, a note asks the user to double-check them, and the prediction is still shown.
- If absences are above 40, a note says such values are rare in the training data, so the prediction may be less reliable.

- The absences field only accepts values from 0 to 75, the range found in the data.

Edge-case tests (all with the Random Forest model):

#	Input	Expected	Actual	Result
1	G1 = 0, G2 = 0	Error, no prediction	Red error: Both G1 and G2 are 0. No prediction shown.	Pass
2	G1 = 18, G2 = 6	Note and a prediction	Blue note about the grade gap; At Risk (59%)	Pass
3	Absences typed as 80	Value rejected	Field marked invalid; value above 75 not accepted	Pass
4	G1 = 2, G2 = 1, absences = 50	Note and a prediction	Blue note about rare absences; At Risk (94%)	Pass
5	G1 = 5, G2 = 4, failures = 2	At Risk	At Risk (95%)	Pass
6	G1 = 15, G2 = 16, failures = 2	Not At Risk	Not At Risk (22%)	Pass

Test 6 shows how strongly the model depends on the earlier grades: even with two past failures, high G1 and G2 gave a low risk estimate.

[Insert screenshot: screenshots/03_validation_error.png]

Figure 9: Validation error for G1 = 0 and G2 = 0.

[Insert screenshot: screenshots/04_validation_note.png]

Figure 10: Validation note for a large gap between G1 and G2.

12. Limitations and Responsible Use

- The dataset covers a mathematics course at two schools (coded GP and MS) and may not represent students at other institutions.
- The risk threshold of 10 out of 20 is a project assumption and must be explained wherever the results are used.
- The test set is small (79 students), so reported metrics are estimates and may change with a different split.

- The model depends mainly on G1 and G2. It works as an early warning after period 2, not as an early prediction before any grades exist.
- Predictions can be wrong and may reflect bias in historical data. Sex and relationship status were removed, but other personal attributes remain in the data, such as parents' education and jobs, address, age and alcohol use. A stricter version could remove them as well.
- Student privacy must be protected, and sensitive attributes should be avoided where they are not needed.
- The prediction should be used as a prompt for supportive intervention, not as a final judgment about ability or future success, and not to punish or label students.

13. Conclusion and Future Work

This project delivered a small end-to-end prototype that accepts selected student features, estimates an academic-risk category, displays basic charts and reports evaluation metrics. The Random Forest achieved an F1-score of 0.87 for the At Risk class on the held-out test set, slightly ahead of the Logistic Regression baseline at 0.83. These figures were measured on this dataset and should not be assumed to hold elsewhere.

Future work could include testing on data from other institutions, comparing models without G1 and G2 to see how early a warning is possible, using cross-validation for more reliable estimates, trying other models, and adding per-student explanations such as SHAP.

14. References

1. P. Cortez and A. Silva, Using Data Mining to Predict Secondary School Student Performance, 2008. UCI Machine Learning Repository: Student Performance dataset.
2. F. Pedregosa et al., Scikit-learn: Machine Learning in Python, Journal of Machine Learning Research, 2011.
3. Streamlit documentation, https://docs.streamlit.io

Appendix: Project Structure and How to Run

```
student-risk-prediction/
    data/           student-mat.csv, student-mat-cleaned.csv
    src/            clean_data.py, data_preprocessing.py, train_model.py,
evaluate_model.py
    models/         saved models and test split
    figures/        confusion matrices and EDA charts
    screenshots/    dashboard screenshots
    app.py          Streamlit dashboard
    requirements.txt
    README.md
```

To run the project:

```
pip install -r requirements.txt
python src/clean_data.py
python src/train_model.py
python src/evaluate_model.py
python -m streamlit run app.py
```